

L04 - CNNs Using MNIST Dataset

Team Contributions

Mary Ann:

For this lab, I focused on changing many of the inputs and comparing the results across different epochs and batch sizes. I doubled the batch size for each run (128 – 256 – 512) and tested both 15 and 25 epochs. With 25 epochs, the larger batch sizes improved more and had the highest accuracy, and the test loss also increased at 25 epochs.

Overall, my architecture ended up with 3 Conv2D layers, three MaxPooling layers, a Flatten layer, a Dropout layer, and a final Dense SoftMax layer. The best accuracy I saw was 0.9915, which I am happy with. It was better than the baseline CNN and had enough of a difference to analyze and compare how the different hyperparameters affect performance.

Cesar:

My individual contribution to the project was to focus on understanding and explaining the Conv2D and MaxPooling layers. I worked on understanding how Conv2D looks for different patterns in images, such as lines, edges, curves, and shapes, and how MaxPooling reduces the size of the information while keeping the important patterns. This part helped me understand better how the CNN processes an image and allowed me to continue understanding the other parts of the model with my group.

Luiz:

My contribution to the project focused on reviewing the CNN notebook and testing different model parameters. I looked at the Conv2D filters and kernel size, the MaxPooling layers, batch size, and number of epochs to understand how they affected the model performance. I compared the original results with different settings and found that using a batch size of 256 and increasing the epochs improved the accuracy. The model reached 99.31% accuracy with 100 epochs. I also helped review the other parts of the CNN, including Dropout, Flatten, Adam optimizer, and categorical cross-entropy, to better understand how they work together in the model.

Nichelle:

My individual technical contribution to this lab was completing and testing the CNN notebook independently and documenting my own observations and results. I reviewed the model architecture, completed the required compilation and training steps, and modified the second Conv2D layer from 64 to 128 filters.

I also conducted a training-parameter experiment by changing the batch size from 128 to 64 and reducing the number of epochs from 15 to 10 while keeping the model architecture the same. I compared the training and validation performance of the runs and evaluated the experimental model on the test

dataset, where it achieved 99.19% test accuracy with a test loss of 0.0253. I completed the notebook reflection questions based on my own results and observations.

In addition to the technical work, I handled much of the administrative and organizational work for the group submission. I reviewed the assignment requirements and the materials contributed by the team members, helped clarify how the notebook, reflections, and individual contributions should be organized, and worked to consolidate the group's documentation into a consistent submission. I also organized the final files, checked that the required components were included, and prepared the materials for placement in the team GitHub repository and final submission.